

EUTRA/LTE NB-IoT MEASUREMENT APPLICATION

Specifications

R&S®FSW3-KM313

R&S®FSW-K106

R&S®FSV3-K106

R&S®FPS-K106

R&S®VSE-K106



Specifications
Version 05.00

ROHDE & SCHWARZ

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Definitions

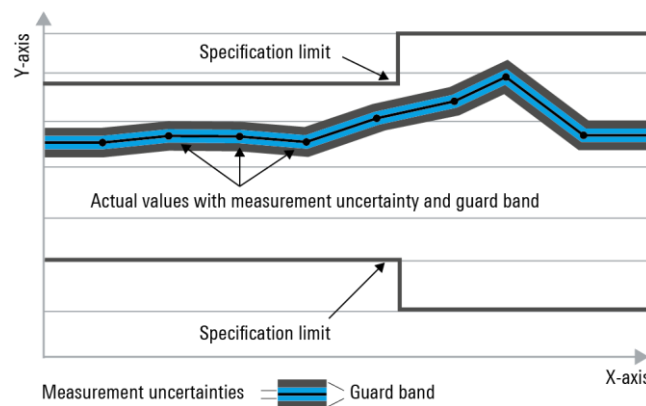
General

Product data applies under the following conditions:

- Three hours of storage at ambient temperature followed by 30 minutes of warm-up operation
- Specified environmental conditions met
- Recommended calibration interval adhered to
- All internal automatic adjustments performed, if applicable

Specifications with limits

Represent warranted product performance by means of a range of values for the specified parameter. These specifications are marked with limiting symbols such as $<$, $\leq$, $>$, $\geq$, $\pm$ or descriptions such as maximum, limit of, minimum. Compliance is ensured by testing or is derived from the design. Test limits are narrowed by guard bands to take into account measurement uncertainties, drift and aging, if applicable.



Non-traceable specifications with limits (n. trc.)

Represent product performance that is specified and tested as described under "Specifications with limits" above. However, product performance in this case cannot be warranted due to the lack of measuring equipment traceable to national metrology standards. In this case, measurements are referenced to standards used in the Rohde & Schwarz laboratories.

Specifications without limits

Represent warranted product performance for the specified parameter. These specifications are not specially marked and represent values with no or negligible deviations from the given value, e.g. dimensions or resolution of a setting parameter. Compliance is ensured by design.

Typical data (typ.)

Characterizes product performance by means of representative information for the given parameter. When marked with $<$, $>$ or as a range, it represents the performance met by approximately 80 % of the instruments at production time. Otherwise, it represents the mean value.

Nominal values (nom.)

Characterize product performance by means of a representative value for the given parameter, e.g. nominal impedance. In contrast to typical data, a statistical evaluation does not take place and the parameter is not tested during production.

Measured values (meas.)

Characterize expected product performance by means of measurement results gained from individual samples.

Uncertainties

Represent limits of measurement uncertainty for a given measurand. Uncertainty is defined with a coverage factor of 2 and has been calculated in line with the rules of the Guide to the Expression of Uncertainty in Measurement (GUM), taking into account environmental conditions, aging, wear and tear.

Device settings and GUI parameters are designated with the format "parameter: value".

Non-traceable specifications with limits, typical data as well as nominal and measured values are not warranted by Rohde & Schwarz.

In line with the 3GPP standard, chip rates are specified in million chips per second (Mcps), whereas bit rates and symbol rates are specified in billion bit per second (Gbps), million bit per second (Mbps), thousand bit per second (kbps), million symbols per second (Msps) or thousand symbols per second (ksps), and sample rates are specified in million samples per second (Msample/s). Gbps, Mcps, Mbps, Msps, kbps, ksps and Msample/s are not SI units.

Specifications

The specifications of the R&S®FSW3-KM313, R&S®FSW-K106, R&S®FSV3-K106, R&S®FPS-K106 and R&S®VSE-K106 are based on the specifications of the FSWX, FSW, R&S®FSVA3000, R&S®FSV3000, R&S®FPS, FPL signal and spectrum analyzers and the R&S®RTO6 oscilloscope. They have not been checked separately and are not verified during instrument calibration. Measurement uncertainties are given as 95 % confidence intervals. The specified level measurement errors do not take into account systematic errors due to reduced signal-to-noise ratio (SNR).

General remarks

This specifications document covers the R&S®FSW3-KM313, R&S®FSW-K106, R&S®FSV3-K106, R&S®FPS-K106 and the R&S®VSE-K106 EUTRA/LTE NB-IoT measurement applications.

The R&S®FSW3-KM313, R&S®FSx-K106 and the R&S®FPS-K106 run on the analyzer itself.

The R&S®VSE-K106 runs on a PC that can be connected to analyzers and oscilloscopes as specified below.

Overview

Assignment of option numbers to link modes and access modes

R&S®FSW3-KM313	EUTRA/LTE NB-IoT downlink
R&S®FSW-K106	EUTRA/LTE NB-IoT downlink
R&S®FSV3-K106	EUTRA/LTE NB-IoT downlink
R&S®FPS-K106	EUTRA/LTE NB-IoT downlink
R&S®VSE-K106	EUTRA/LTE NB-IoT downlink and uplink, this application runs on the devices FSWX, FSW, R&S®FSVA3000, R&S®FSV3000, R&S®FPS and FPL

Minimum system requirements for R&S®VSE vector system explorer software

Operating system	Windows 10, 64 bit/Windows 11, 64 bit, version 24H2 and 25H2
Hard disk space	5 Gbyte required
RAM	≥ 4 Gbyte required
CPU	1.5 GHz minimum (> 2.5 GHz recommended)
Graphics resolution	≥ 1280 × 1024 pixel
USB	1 free USB port for connecting the R&S®FSPC smart card reader (if no PC built-in smart card reader is used)
Measuring instrument connection	LAN connection (VXI-11, Hi-Slip), VISA driver

EUTRA/LTE analysis

Signal acquisition

Supported standards	downlink	EUTRA/LTE NB-IoT in line with 36.211 Rev. 13.2.0 36.104 Rev. 15.4.0
	uplink	EUTRA/LTE NB-IoT in line with 36.211 Rev. 13.2.0 36.101 Rev. 14.0.0
	downlink	2.01 ms to 50.1 ms
	uplink FSWX, FSW, R&S®FSVA3000, R&S®FSV3000, R&S®FPS	2.01 ms to 20.48 s
Capture length	uplink FPL	2.01 ms to 12 s
	uplink R&S®RTO6	2.01 ms to 20.48 s
Trigger modes		free run, external

Measurement parameters downlink

	R&S®FSW3-KM313, R&S®FSW-K106, R&S®FSV3-K106, R&S®FPS-K106, R&S®VSE-K106	
Deployment		in-band, guard-band, stand-alone
Input		RF, file
Channel bandwidth		200 kHz
MIMO configuration		1, 2 TX antennas
Number of input channels		1
NPDSCH resource allocation detection		auto
Switchable multicarrier filter		•
Adjacent channel leakage power ratio (ACLR)		
Assumed adjacent channel carrier		stand-alone NB-IoT

Measurement parameters uplink

		R&S®VSE-K106
Deployment		stand-alone
Input		RF, file
Subcarrier spacing		3.75 kHz, 15 kHz
NPUSCH format		1, 2
Number of subcarriers		1, 3, 6, 12
Modulation types		BPSK, QPSK
Automatic NPUSCH detection		number of subcarriers, start subcarrier, modulation type
Adjacent channel leakage power ratio (ACLR)		
Assumed adjacent channel carrier		GSM, UTRA

Result displays downlink

	R&S®FSW3-KM313 R&S®FSW-K106 R&S®FSV3-K106 R&S®FPS-K106	R&S®VSE-K106
Result summary		
Center frequency error	•	•
Sampling error	•	•
RSTP (reference signal TX power)	•	•
OSTP (OFDM symbol TX power)	•	•
RSSI (received signal strength indicator)	•	•
Power	•	•
Crest factor	•	•
Power versus time		
Capture buffer	•	•
Power versus symbol and carrier	•	•
EVM		
EVM versus carrier	•	•
EVM versus symbol	•	•
EVM versus symbol and carrier	•	•
Frequency error versus symbol	•	•
Spectrum		
Power spectrum	•	•
Channel flatness	•	•
Channel flatness difference	•	•
ACLR	•	•
Spectrum emission mask	•	•
Constellation		
Constellation diagram	•	•
Statistics/miscellaneous		
CCDF	•	•
Group delay	•	•
Allocation summary list	•	•
Allocation ID versus symbol and carrier	•	•
Time alignment error	•	•

Result displays uplink

	R&S®VSE-K106
Result summary	
3GPP EVM NPUSCH BPSK	•
3GPP EVM NPUSCH QPSK	•
3GPP EVM NDRMS NPUSCH BPSK	•
3GPP EVM NDRMS NPUSCH QPSK	•
3GPP frequency error	•
EVM all	•
EVM physical channel	•
EVM physical signal	•
Center frequency error	•
I/Q offset	•
I/Q gain imbalance (only for 12 carrier signals)	•
I/Q quadrature error (only for 12 carrier signals)	•
Power	•
Crest factor	•
Power versus time	
Capture buffer	•
Power versus symbol and carrier	•
EVM	
EVM versus carrier	•
EVM versus symbol	•
EVM versus symbol and carrier	•
Spectrum	
Power spectrum	•
Spectrum flatness	•
Spectrum flatness difference	•
Group delay	•
ACLR	•
Constellation	
Constellation diagram	•
Statistics/miscellaneous	
CCDF	•
Bitstream	•
Allocation summary list	•

Measurement uncertainty (nominal)

Specifications apply under the following conditions: temperature from +20 °C to +30 °C; signal level –25 dBm; properly adjusted reference level; external reference frequency applied, frequency range from 700 MHz to 1.98 GHz

Base station and UE output power

Base station and UE output power	FSW	R&S®FSVA3000	R&S®FSV3000	R&S®FPS	FPL	R&S®RTO6
Level uncertainty	same as FSW/R&S®FSVA3000/R&S®FSV3000/R&S®FPS/FPL/R&S®RTO6 (see FSW/R&S®FSVA3000/R&S®FSV3000/R&S®FPS/FPL/R&S®RTO6 total measurement uncertainty)					

Transmitted signal quality base station

EVM	FSW	R&S®FSVA3000	R&S®FSV3000	R&S®FPS	FPL	R&S®RTO6
Residual EVM	< 0.33 % (–49.8 dB)	< 0.34 % (–49.4 dB)	< 0.35 % (–49.1 dB)	< 0.47 % (–46.6 dB)	< 0.36 % (–48.9 dB)	< 1.00 % (–40.0 dB)

Transmitted signal quality UE

EVM		FSW	R&S®FPS	FPL	R&S®RTO6
Residual EVM	15 kHz carrier spacing 12/6/3 carrier signals	< 0.12 % (–58.5 dB)	< 0.20 % (–54.0 dB)	< 0.22 % (–53.3 dB)	< 0.40 % (–48.0 dB)
	15 kHz carrier spacing 1 carrier signal	< 0.02 % (–74.0 dB)	< 0.10 % (–60.0 dB)	< 0.10 % (–60.0 dB)	< 0.20 % (–54.0 dB)
	3.75 kHz carrier spacing	< 0.02 % (–74.0 dB)	< 0.18 % (–54.9 dB)	< 0.20 % (–54.0 dB)	< 0.25 % (–52.0 dB)

Ordering information

Designation	Type	Order No.
EUTRA/LTE NB-IoT measurement applications		
EUTRA/LTE NB-IoT downlink measurement application	R&S®FSW3-KM313	3706.8573.02
EUTRA/LTE NB-IoT downlink measurement application	R&S®FSW-K106	1331.6351.02
EUTRA/LTE NB-IoT downlink measurement application	R&S®FSV3-K106	1346.3418.02
EUTRA/LTE NB-IoT downlink measurement application	R&S®FPS-K106	1331.3246.02
EUTRA/LTE NB-IoT downlink and uplink measurement application	R&S®VSE-K106	1320.7900.02
Analyzers		
FSWX		
Signal and spectrum analyzer, 2 Hz to 26.5 GHz, 1 RF	R&S®FSWX3026	1348.0000.21
Signal and spectrum analyzer, 2 Hz to 26.5 GHz, 2 RF	R&S®FSWX3026	1348.0000.22
Signal and spectrum analyzer, 2 Hz to 44 GHz, 1 RF	R&S®FSWX3044	1348.0000.41
Signal and spectrum analyzer, 2 Hz to 44 GHz, 2 RF	R&S®FSWX3044	1348.0000.42
FSW		
Signal and spectrum analyzer, 2 Hz to 8 GHz	R&S®FSW8	1331.5003.08
Signal and spectrum analyzer, 2 Hz to 13.6 GHz	R&S®FSW13	1331.5003.13
Signal and spectrum analyzer, 2 Hz to 26.5 GHz	R&S®FSW26	1331.5003.26
Signal and spectrum analyzer, 2 Hz to 43.5 GHz	R&S®FSW43	1331.5003.43
Signal and spectrum analyzer, 2 Hz to 50 GHz	R&S®FSW50	1331.5003.50
Signal and spectrum analyzer, 2 Hz to 67 GHz	R&S®FSW67	1331.5003.67
Signal and spectrum analyzer, 2 Hz to 85 GHz	R&S®FSW85	1331.5003.85
R&S®FSVA3000, R&S®FSV3000		
Signal and spectrum analyzer, 10 Hz to 4 GHz	R&S®FSVA3004	1330.5000.05
Signal and spectrum analyzer, 10 Hz to 7.5 GHz	R&S®FSVA3007	1330.5000.08
Signal and spectrum analyzer, 10 Hz to 13.6 GHz	R&S®FSVA3013	1330.5000.14
Signal and spectrum analyzer, 10 Hz to 30 GHz	R&S®FSVA3030	1330.5000.31
Signal and spectrum analyzer, 10 Hz to 44 GHz	R&S®FSVA3044	1330.5000.44
Signal and spectrum analyzer, 10 Hz to 4 GHz	R&S®FSV3004	1330.5000.04
Signal and spectrum analyzer, 10 Hz to 7.5 GHz	R&S®FSV3007	1330.5000.07
Signal and spectrum analyzer, 10 Hz to 13.6 GHz	R&S®FSV3013	1330.5000.13
Signal and spectrum analyzer, 10 Hz to 30 GHz	R&S®FSV3030	1330.5000.30
Signal and spectrum analyzer, 10 Hz to 44 GHz	R&S®FSV3044	1330.5000.43
R&S®FPS		
Signal and spectrum analyzer, 10 Hz to 4 GHz	R&S®FPS4	1319.2008.04
Signal and spectrum analyzer, 10 Hz to 7 GHz	R&S®FPS7	1319.2008.07
Signal and spectrum analyzer, 10 Hz to 13.6 GHz	R&S®FPS13	1319.2008.13
Signal and spectrum analyzer, 10 Hz to 30 GHz	R&S®FPS30	1319.2008.30
Signal and spectrum analyzer, 10 Hz to 40 GHz	R&S®FPS40	1319.2008.40

Designation	Type	Order No.
FPL		
Spectrum analyzer, 5 kHz to 3 GHz	R&S®FPL1003	1304.0004.03
Spectrum analyzer, 5 kHz to 7.5 GHz	R&S®FPL1007	1304.0004.07
Spectrum analyzer, 5 kHz to 14 GHz	R&S®FPL1014	1304.0004.14
Spectrum analyzer, 5 kHz to 26.5 GHz	R&S®FPL1026	1304.0004.26
Spectrum analyzer, 10 Hz to 44 GHz	R&S®FPL1044	1304.0004.44
Oscilloscope		
Base unit, 200 Mpoints/800 Mpoints, 4 channels, bandwidth option required	R&S®RTO64	1802.0001.04
I/Q software interface	R&S®RTO6-K11	1801.6812.02
Vector signal explorer software		
Basic edition	R&S®VSE	1345.1011.06
Enterprise edition	R&S®VSE	1345.1105.06
R&S®VSE software maintenance	R&S®VSE-SWM	1320.7622.81

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